

**Course Title: CONSTRUCTION TECHNOLOGY AND
PROJECT MANAGEMENT (CE351)**

B.Tech Civil Engineering: Third year, 2nd Sem
Year – 2025-26

Course Instructor: Dr. Deepak M.D.

INTRODUCTION

Project: Project is defined as “a temporary endeavor undertaken to create a unique product, service or result” [p. 434 PMI, 2008]. E.g. Residential Project.

Unique Features of Construction Projects

- ☐ One-time activity
- ☐ Complexity
- ☐ High cost and time for execution
- ☐ High risk of failure
- ☐ Difficulty in defining quality standards
- ☐ Uniqueness of people’s relationship
- ☐ Feedback mechanism
- ☐ Lack of experience of client or owner
- ☐ Untrained workforce

Project Categories

Building construction – Includes residential and commercial complexes, educational and recreational facilities, hospitals, hotels etc. Designed by architect/engineering firms and are financed by public and private sector and individuals.

Infrastructure construction – Include capital intensive and heavy-equipment oriented works which involves movement of large quantity of bulk materials like earth, steel and concrete. Includes dams and canals, highways and airports, railways and bridges, large water supply and sewage disposal network, nuclear and thermal power plants etc. which build up the infrastructure for the growth of the economy.

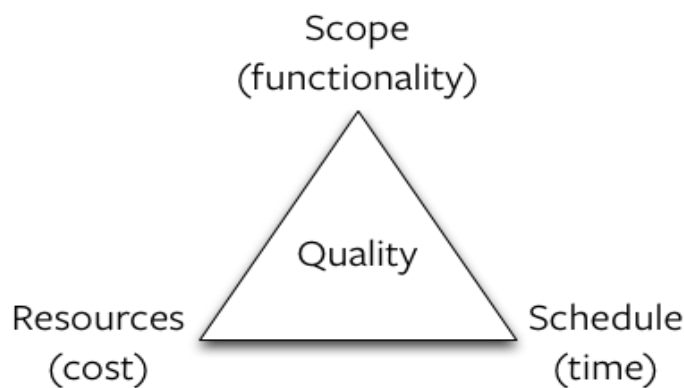
Industrial construction – Includes construction of manufacturing, processing and industrial plants like steel mills, petroleum refineries etc. Involves heavy investment and are highly specialized. These are financed by government, public and private companies.

Project Management

Project Management is the art and science of mobilizing and managing manpower, materials, machineries and money to complete the assigned project work on time within the budgeted costs and specified technical performance standards.

Definition: Time limited, goal directed, major understanding that require the commitment of various skills and resources. It also describes the project as a combination of human and non-human resources pooled together in a temporary organization to achieve specific purpose and set of activities which can achieve that purpose to distinguish one project from the other.

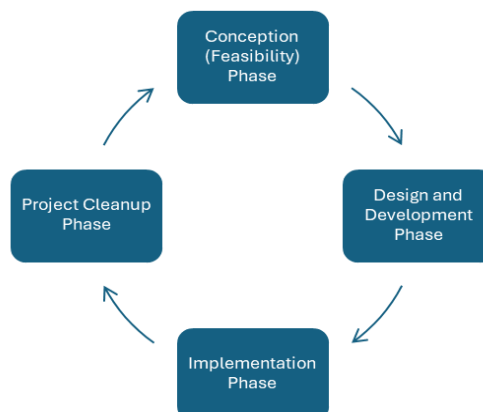
Project Management Triangle



Project Objectives:

- The project should be completed with minimum of elapsed time.
- Project should use the available manpower and other resources as sparingly as possible without any delay.
- Project should be completed with a minimum of capital investment without delay.

Project Life Cycle Phase



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According to the PMBOK Guide (Project Management Body of Knowledge) by the Project Management Institute (PMI), a project management life cycle consists of 5 distinct phases including initiation, planning, execution, monitoring, and closure that combine to turn a project idea into a working product.

The 5 basic phases in the project management process are:

- Project Initiation
- Project Planning
- Project Execution
- Project Monitoring and Controlling
- Project Closing

Role of Project Manager

☐ Figure head role

☐ Disseminator role

☐ Disturbance handling role

☐ Negotiations role

☐ Resources allocator role

☐ Leadership role

☐ Monitoring role

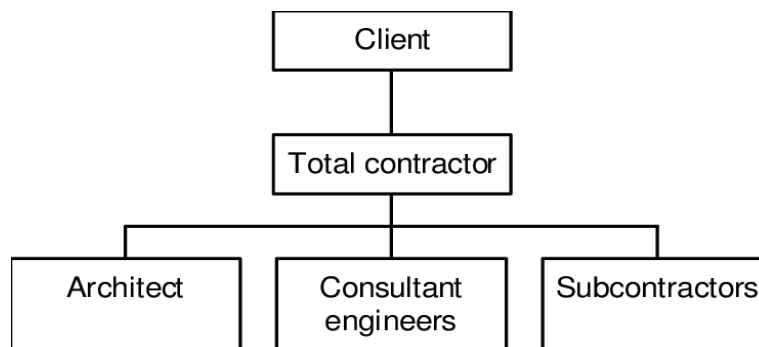
Stakeholders of a Construction Project

- Architect
- Client (Owner)
- Contractor
- Consultant
- Subcontractor/Supplier/Vendor
- Lawyer, Insurer

Organization Breakdown Structure (OBS)

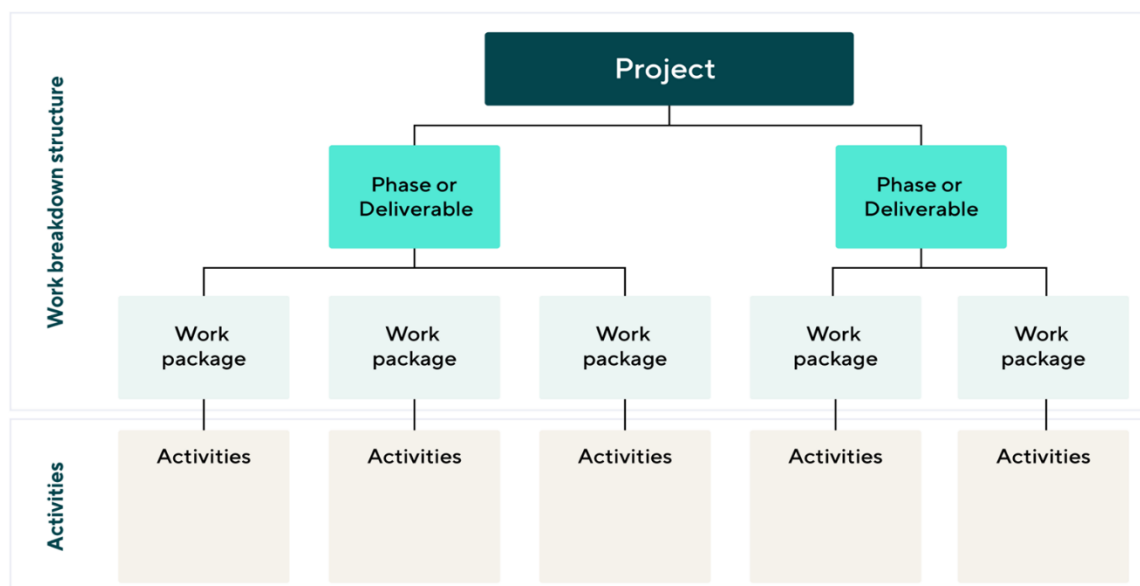
Systematic structure for assigning the responsibilities in an organization.

The organizational structure is about how to use one of the basic resources, people, and how to facilitate overcoming the communication barriers at organizational interfaces. For example, the organizational structure indicates the arrangement of different departments and the division of labour.



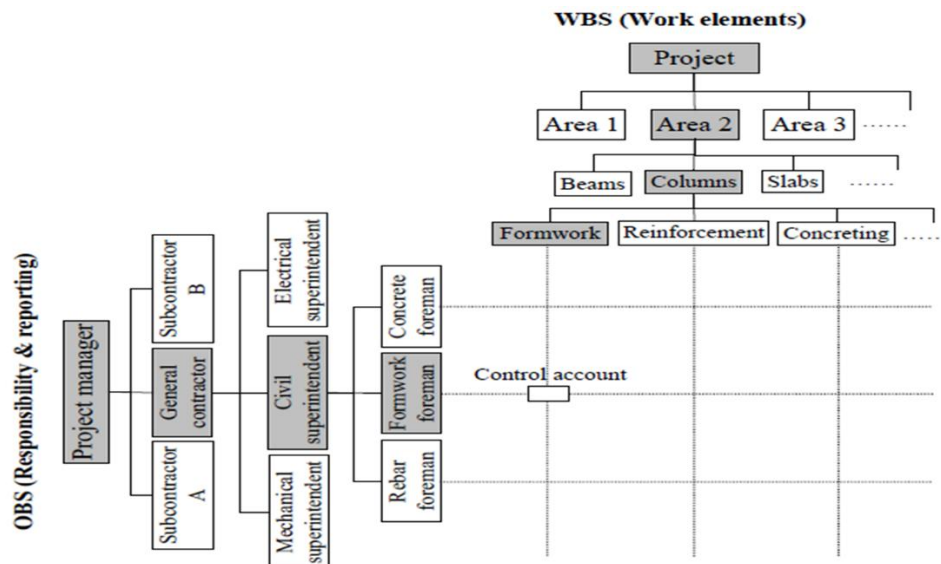
Work Breakdown Structure (WBS)

Work Breakdown Structure (WBS) is an organized, hierarchical representation of all work - broken out in sufficient detail - to support a project's planning, assignment of roles and responsibilities, and ongoing monitoring and control.



Project Responsibility Matrix

Project Responsibility Matrix is a project tracking tool that maps people against specific profiles and tasks in a project. The point is to ensure that as the project progresses, everyone understands who is doing what – and who should be consulted or kept in the loop.



Steps in Project Scheduling

- Identifying the activities
- Estimating durations of activities
- Establishing the logic
- Drawing the schedule
- Analysis of schedule
- Setting baseline schedule
- Updating and revising the schedule

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GANTT / Bar Charts

Bar Chart is a graphical representation of project activities shown in time scaled bar lines with no links shown between the activities.

Advantages of a Gantt Chart

- Simple to prepare and easy to understand
- No 'theory' or complicated calculations are required
- Can be used as a good summary tool – especially at review meetings
- Can be easily changed
- Good for repetitive projects
- Ideal for projects that are simple and sequential

Disadvantages of a Gantt Chart

- Difficult to depict interdependencies clearly
- Cannot establish criticality of activities
- Can be very complex to prepare for large projects
- Construction logic (activity sequencing assumption) is not easily obtained as relationships are lacking

Problem #1

The activity breakdown for a certain project is as shown in Table. Prepare the Bar Chart for the Project.

Activity No.	Duration (weeks)
1	1
2	2
3	4
4	3
5	1
6	2
7	4

- Activity 2 and Activity 3 can be done concurrently, and both must follow Activity 1.
- Activity 2 must precede Activity 4.
- Activity 5 cannot begin until both Activities 2 and 3 are completed.
- Activity 6 can be started only after Activities 4 and 5 are complete.
- Activity 7 is the last Activity which can be started only after completion of Activity 5.

Problem #2

For the following activities, determine the duration for completion of the project.

Activities	A	B	C	D	E	F	G	H	I	J
Preceding activity	None	None	A & B	A	C	A & B	F	G	G	I
Time in days	40	20	15	25	15	25	10	15	10	15

Project Scheduling using Networks

Network Diagram: Graphical flow plan of the activities that must be accomplished for completing the project.

It is the arrangement of the activities in a logical sequence, following the precedence-succedence relationships between the activities.

There are three popular network-scheduling techniques. They are:

- Critical Path Method (CPM)
- Precedence Diagramming Method (PDM)
- Programme Evaluation Review Technique (PERT)

Development of a network

First tabulate the network logic and then drawing the arrow diagram step by step.

The logic of activity can be made by questioning each activity are as follows:

- Which activity/activities precede this activity?
- Which activity/activities succeed or follow this activity?
- Are there any logical constraints?
- Is it the final activity?

Types of Networks

Two kinds of networks: AOA & AON

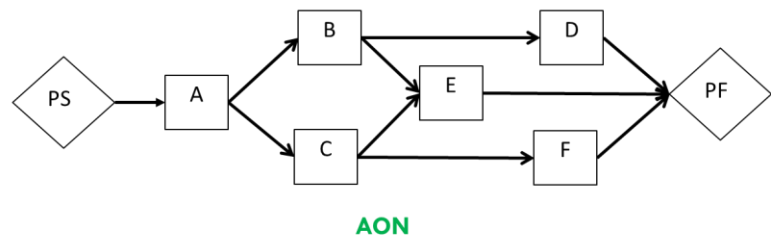
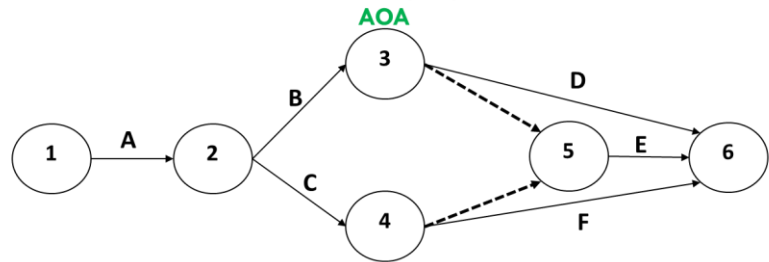
AOA (Activity on Arrow technique) - Activities are shown as arrows leading from one node to another and the nodes represent the start or end of an activity (Events).

AON (Activity on Node technique) - Activities are denoted by circles or boxes called nodes, and the immediate predecessor relationship between the two activities is shown by an arrow connecting the two nodes.

Problem #1

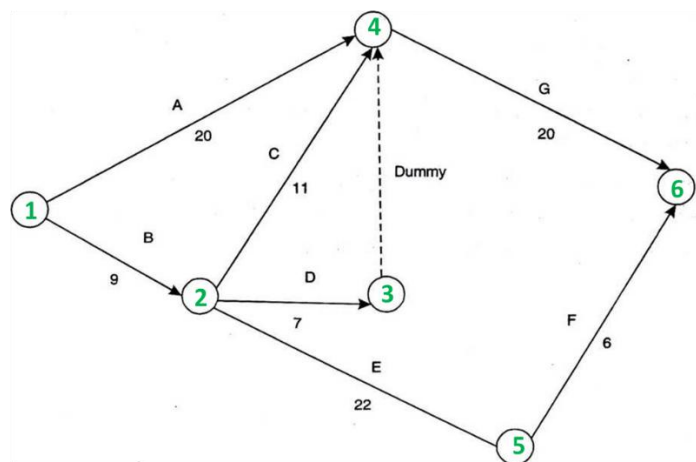
Draw the arrow network (AOA and AON) for the project given:

Activity	Preceding Activity
A	-
B	A
C	A
D	B
E	B,C
F	C



Problem #2

Activity	Duration (days)	Preceding Activity
A	20	-
B	9	-
C	11	B
D	7	B
E	22	B
F	6	E
G	20	A,C,D



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Problem #3

Activity	Duration (days)	Predecessor
A	10	-
B	9	-
C	12	-
D	12	A
E	4	A
F	8	C
G	20	D
H	3	D
I	9	B, E, F
J	13	C
K	6	G
L	18	H, I, M
M	7	J
N	9	J

**Draw the
network diagram
(AOA) for the
given project**

Critical Path Method (CPM)

Critical Path Method (CPM) is a procedure for using network analysis to identify those tasks which are on the critical path.

For all tasks off the critical path, a degree of tolerance is possible (e.g. early start, early completion, late start, late completion, etc.).

A network represents the logical sequence of activities contained in each path and activities in a network will have a different duration.

The path that has the longest duration is called the critical path and the activities that lie on the critical path are called critical activities. It is the critical path that sets the overall duration of the project. The critical path determines the earliest possible time in which the project can be completed.

The study of non-critical activities with regard to the 'free time' available is Activity Float Analysis. This helps in better control of project implementation and better allocation of resources. The free time available for an activity is called 'float'.

Types of Floats:

- Total Float
- Free Float
- Independent Float
- Interfering Float

Total Float in an activity is the amount of time by which the start of the activity may be delayed without causing a delay in the completion of the project.

Free float of an activity is the amount of time by which the start of the activity may be delayed without delaying the start of the following activity.

Independent float is the amount of time by which the start of an activity may be delayed without affecting the preceding or the following activity.

Interfering float is the difference between the total float and free float of an activity i.e. head event slack. Interfering float is rarely used in construction.

Problem #4

Activity	IPA	Duration (days)
A	-	5
B	A	8
C	A	6
D	B	9
E	B,C	6
F	C	3
G	D,E,F	1

- Draw a neat network diagram.
- Determine the Total float, Free float, Independent float and Interfering float of all activities.
- Locate the critical path

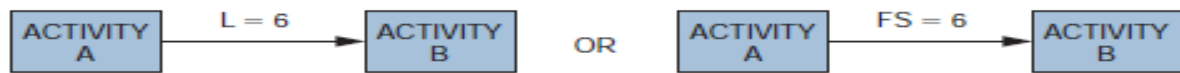
Problem #5

Activity	Duration (days)	Predecessor
A	5	-
B	4	-
C	7	A
D	7	B
E	8	B
F	5	B
G	8	C,D
H	11	E,I
I	2	F
J	12	F
K	11	H,J

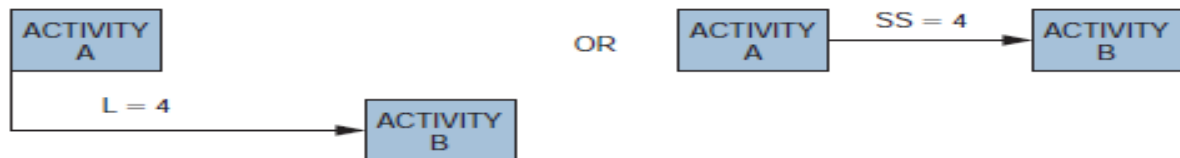
Draw AON network and compute the Total float, Free float, Independent float and Interfering float of all activities.

Precedence Diagram Method (PDM)

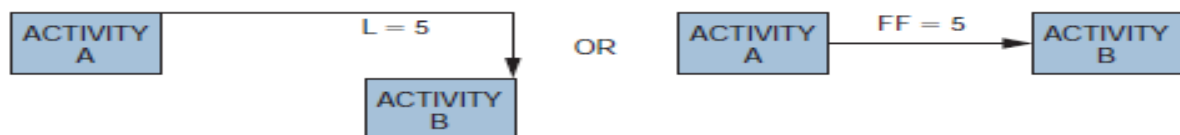
A project management technique using nodes (boxes) for activities and arrows for logical relationships (dependencies) to create a network diagram, visually mapping project flow and sequencing tasks. Also known as Activity-on-Node (AON), PDM defines four dependency types (Finish-to-Start, Start-to-Start, Finish-to-Finish, Start-to-Finish) to show how tasks link, allowing for complex scheduling, overlap, and identifying the critical path for project completion.



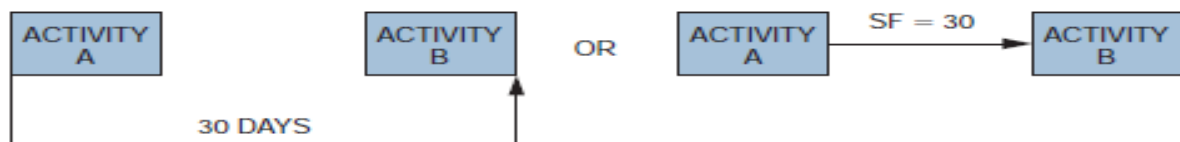
(A) FINISH-TO-START (FS) RELATIONSHIP. THE START OF B MUST LAG 6 DAYS AFTER THE FINISH OF A.



(B) START-TO-START (SS) RELATIONSHIP. THE START OF B MUST LAG 4 DAYS AFTER THE START OF A.



(C) FINISH-TO-FINISH (FF) RELATIONSHIP. THE FINISH OF B MUST LAG 5 DAYS AFTER THE FINISH OF A.



(D) START-TO-FINISH (SF) RELATIONSHIP. THE FINISH OF B MUST LAG 30 DAYS AFTER THE START OF A.

Problem #6

Activity	Preceding activity	Duration
A	-	5
B	A (FS+4)	7
C	A	6
D	B	7
E	C(SS+2), B	10
F	A (SF+15)	13
G	D(FF+4)	3
H	E,F,G	8

Draw PDM network diagram. Find ES,EF, LS,LF and indicate the critical path.

Comparison between AON and PDM

AON	PDM
Allows only one kind of logical relationship between activities.	Allows four kind of logical relationship between activities (FS,SS,SF,FF)
The preceding activity must be completed before any succeeding activity can begin (lag=0)	The preceding activity can have lag (zero, +ve, -ve) between succeeding activity
Cannot create complex schedule	Can be used as flexible scheduling tool
Easy to interpret	Not easy to interpret

Problem #7

Activity Name	Successor	Duration
Start	A	-
A	B,C	2
B	D,E,H	4
C	H	2
D	F (FF-2)	8
E	G (SS+1), F, I	2
F	I	2
G	I	4
H	I	4
I	Finish	6

Draw PDM network diagram and find ES,EF, LS,LF and indicate the critical path.

Programme Evaluation Review Technique (PERT)

Uncertainty in Project Schedules: CPM does not directly model uncertainty – No probability-based query can be modelled in CPM. Uncertainty is an inherent characteristic of all projects. Construction projects have a relatively lower level of uncertainty compared to research/Défense/space projects. Modelling and managing duration uncertainty and risk are important parts of project planning and control. PERT was developed to address the needs of projects being done for the first time = a challenge in estimating activity duration.

The PERT (Program Evaluation and Review Technique) approach addresses situations where uncertainties must be considered. It is an event-oriented technique. It means the focus is upon the start or completion of events rather than activities. Activities that take place between events are not specified. To compute the critical path in PERT, a probabilistic network is converted into a single-time, deterministic CPM model. PERT studies focus on the implications of uncertainties for project time scheduling and event slack by employing statistical tools.

Estimation of the activity duration:

Three key values are used in the time estimate for each activity:

a = optimistic time, which means that the minimum time required for an activity if everything goes as per the ideal condition

m = most likely time, which will be the most probable time for an activity if normal conditions prevail

b = pessimistic time, which means that the time it might take to complete if everything went wrong and an abnormal situation prevails

Estimation of Mean and SD:

The expected or mean time is given by: $Expected\ Time = \frac{(a+4m+b)}{6}$

Variance is how far a set of random numbers is from the mean; the greater the variance, the greater the spread of the curve. $V = \frac{(b-a)^2}{36}$

The standard deviation is a measure of how concentrated the data are around the mean and serves as a measure of uncertainty. $\sigma = \sqrt{V} = \frac{(b-a)}{6}$

Problem #8

Activity	Proceeding Activity	Optimistic time (weeks) t_o	Most likely time (weeks) t_m	Pessimistic time (weeks) t_p
A	-	4	5	6
B	-	4	6	8
C	A	3	5	7
D	B	1	3	5
E	C,D	2	5	8
F	C,D	3	4	11
G	E	4	5	12
H	E	4	4	10
I	F,H	5	6	13

Draw the project network and estimate the total project duration.

Problem #9

Activity	IP	a	m	b
A: Prepare initial design	-	1	3	5
B: Identify new potential clients	-	4	5	12
C: Develop prospectus for tenants	A	2	3	10
D: Prepare final design	A	1	8	9
E: Obtain planning permission	D	1	2	3
F: Obtain finance from bank	E	1	3	5
G: Select contractor	D	2	4	6
H: Construction	G, F	10	17	18
I: Finalize tenant contracts	B, C, E	6	13	14
J: Tenants move in	I, H	1	2	3

1. Schedule the project.

2. What is the probability of completing the project in 36 weeks?

Problem #10

Activity	Predecessor	Optimistic	Likely	Pessimistic
A	-	10	16	20
B	A	7	10	20
C	A	5	7	8
D	B	15	18	21
E	C,B	25	30	32
F	D	6	9	12
G	E,D	21	25	28
H	F,G	6	8	9

Determine:

1. Most Likely Project Duration?
2. Critical path?
3. $P(PD \geq 95d) = ?$
4. $P(PD \leq 85d) = ?$
5. $P(85d \leq PD \leq 95d) = ?$
6. $P(PD \leq ?) = 0.90$

Project Resource Scheduling

Decisions on Resources

- Materials: Periodic order quantities / Storage capacities
- Manpower: Adequate requirement of manpower / Availability of skilled manpower
- Machinery: Specialized machinery / Equipment sharing
- Money: Adequate cash flows / Credit planning

Resource Levelling – Time-limited resource scheduling

In resource levelling, the constraint is the fixed project duration. In this case, the project must be completed by a fixed date. The focus is to reduce peak resource requirements and to smooth out period-to-period assignments. This is also known as “time-limited resource considerations”. Assessment is done by a resource loading chart or a resource histogram. General assumption: Not to consider the activities in the critical path, but focus on non-critical activities.

Problem #11

Activity	Preceding activity	Duration (weeks)	Labour Requirement
A	-	1	2
B	-	2	1
C	-	2	1
D	A	1	1
E	B	3	1
F	C	2	1
G	D	1	1

The project has 7 activities. The project duration should not exceed beyond the minimum project schedule. Only 3 labour are available for the project execution. Find the resource schedule with respect to labour.

Draw AOA network.

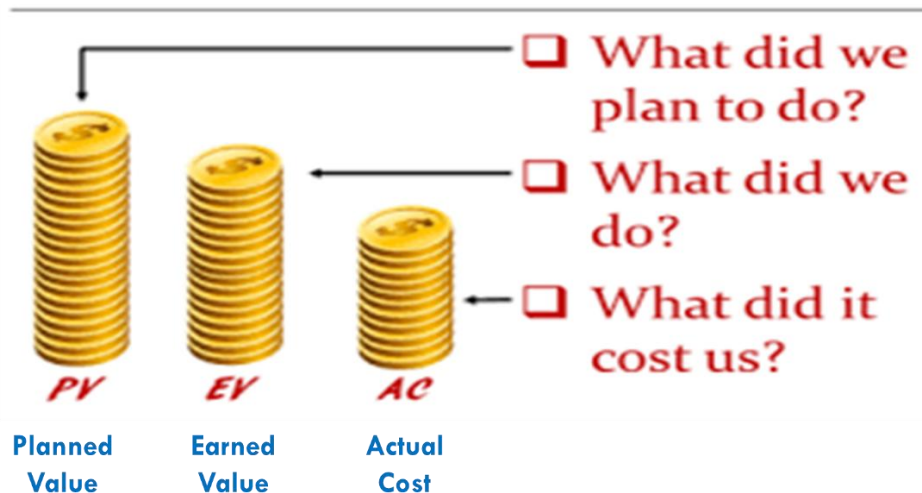
Problem #12

Activity	Preceding activity	Duration (days)	Labour Requirement
A	-	4	7
B	A	3	7
C	A	2	2
D	B	6	5
E	C	4	2
F	D, E	4	2

The details of a small construction project consisting of activities, relationships, and manpower requirements are provided. Manpower availability on any day of the project is 7. Arrive at a schedule of activities that optimizes the manpower requirement subjected to the condition that the project implementation time should not exceed the minimum possible time.

Project Monitoring and Cost Control: Earned Value Management

It is a way to measure the amount of work actually performed on a project. The Earned Value Method integrates schedule and cost control. It is a method for determining the cost, schedule, and technical performance of a project by comparing it with the planned or budgeted performance. It is a way to tell how well a project is “performing” compared to its original plan. Given this information, it is a way to forecast how well the project will perform in the future.

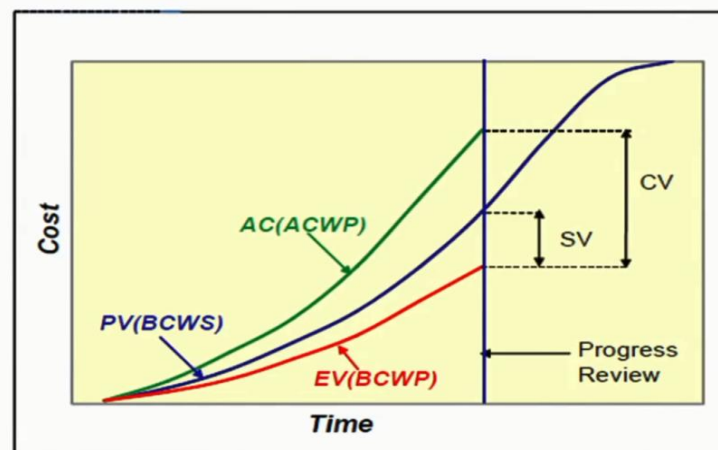


Planned Value (PV) or Budgeted Cost of Work Scheduled (BCWS): It is the cumulative planned value of the authorized work, scheduled for completion to date at contract rate.

Earned Value (EV) or Budgeted Cost of Work Performed (BCWP) : It is the cumulative value of the authorized work completed to date at contract rate.

Actual Cost (AC) or Actual Cost of Work Performed (ACWP): It is the cumulative actual expenses incurred in the project to date.

Project Performance



Source: BIS 15883:2013

Cost Variance (CV): $BCWP - ACWP$

It compares value of work completed vs. what is actually spent.
It helps to identify whether the project is under or over budget.

Negative value indicates that the project is over budget.

CV
< 0 Over Budget
> 0 Under Budget
=0 On Budget

Schedule Variance (SV): $BCWP - BCWS$

It compares value of work completed vs. what is actually planned.

It helps to identify whether the project is ahead or behind the schedule.

Negative value indicates that the project is behind schedule.

SV
< 0 Behind Schedule
> 0 Ahead of Schedule
=0 On Schedule

Cost Performance Index (CPI): $BCWP / ACWP$

It helps to identify whether the project is over or under budget.

Less than 1 means the project is over budget.

CPI
< 1 Over Budget
> 1 Under Budget
= 1 On Budget

Schedule Performance Index (SPI): $BCWP / BCWS$

It helps to identify whether the project is ahead or behind of schedule.

Less than 1 indicates that the project is behind schedule.

SPI
< 1 Behind Schedule
> 1 Ahead of Schedule
=1 On Schedule

Estimate at Completion (EAC):

$EAC = BAC / CPI$ or

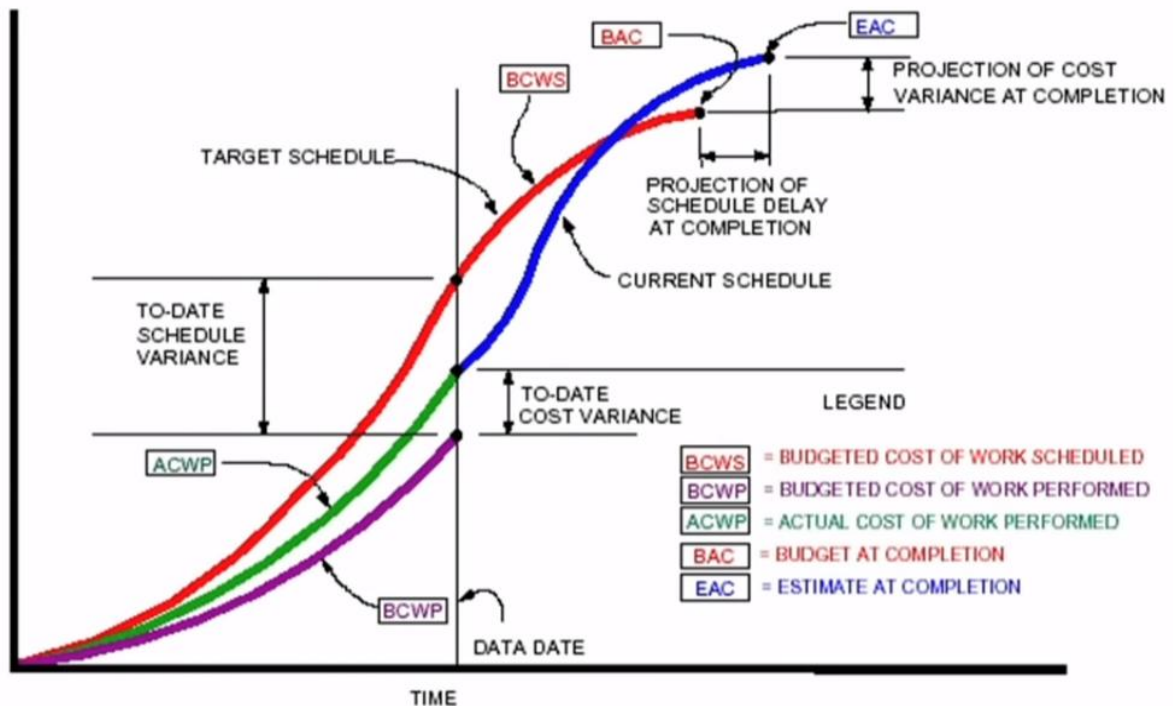
$EAC = AC + [(BAC - EV) / CPI * SPI]$

It is calculated as the actual cost to date plus an appropriate estimate of costs for remaining work.

There are several ways to compute EAC.

Variance at Completion (VAC): $BAC - EAC$

A positive value of VAC means the forecast is for an 'under-run', while a negative VAC means 'over-run' for the project.



Problem #13

A contractor has bagged a fixed cost contract for the supply, installation, testing, and commissioning of 200 solar panels of the same specification at a cost of Rs. 600 lakhs. The contractor had estimated an output of 10 panels per day, so the entire work will be done in 20 days. After completion of 16 days, the project status is reviewed. At the time of review, only 120 panels had been installed, and the cost incurred was Rs. 380 lakhs. It was estimated that Rs. 260 lakhs would be required to complete the pending work, i.e., the installation of the remaining 80 panels. Make a project performance analysis.

Problem #14

A contractor has agreed to install 30 units (electrical fixtures) in 90 days at a price of Rs. 800 per unit. Twenty days later, the contractor has finished 8 units, with an actual total cost (including overhead and profit) of Rs. 6,800. What is the status of the project?

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Problem #15

Calculate the cost variance (CV), the cost performance index (CPI), the schedule variance (SV), and the schedule performance index (SPI), and analyse the cost and schedule performance for each of the projects.

Project	BCWS	BCWP	ACWP	CV	CPI	Cost analysis	SV	SPI	Schedule analysis
(1)	(2)	(3)	(4)	(5) = (3) - (4)	(6) = (3)/(4)	(7)	(8) = (3) - (2)	(9) = (3)/(2)	(10)
1	50,000	50,000	50,000	0	1	On target	0	1	On target
2	50,000	40,000	60,000	-20,000	0.67	Over budget	-10,000	0.8	Behind schedule
3	40,000	40,000	50,000						
4	50,000	50,000	40,000						
5	40,000	50,000	50,000						
6	60,000	40,000	60,000						

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Project Updating: Project Updating can be defined as the planning and programming of the remaining portion of an activity job, incorporating the latest available information.

The updating frequency depends on the nature and type of the project and the contractual provisions. It could be weekly, monthly, or even quarterly. The updating frequency may be lower at the start of the project, while more frequent updates would be needed towards the project's completion.

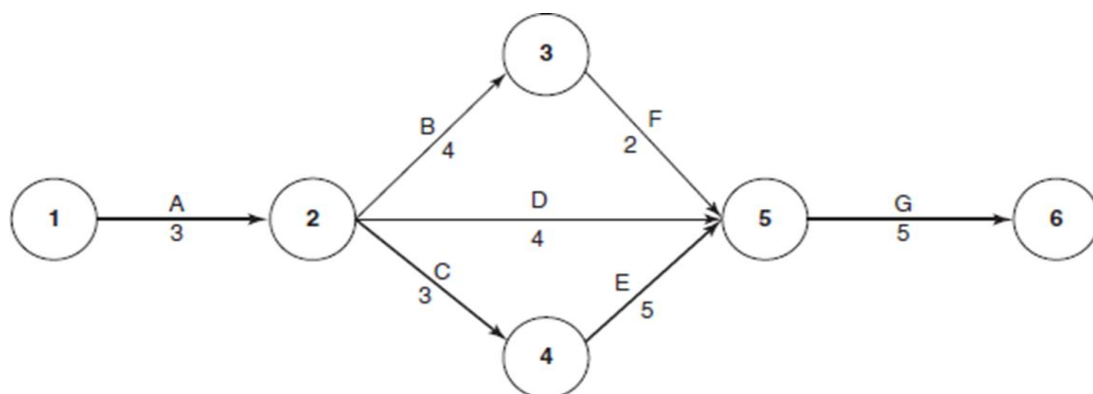
- Updating using Bar Chart
- Updating using PERT/CPM
- Updating using Precedence Network

Project Updating Rules:

The following steps are suggested:

- All completed activities are given zero duration.
- All activities in progress are given a duration equal to the number of days remaining for completion.
- All activities not yet started are given a duration as reported in the latest estimate.
- Any correlation change, i.e., changed network logic between activities yet to be completed, is brought forth, and necessary changes in the arrow network diagram are duly discussed and communicated/recorded.
- To keep up the calendar scheduling of the time baseline unchanged and also keep the facts as they are, the earliest occurrence time of the first activity of the network being updated is marked as, say, X. Where X is the day on which the network is being updated.
- New critical paths are then computed for further control/monitoring.

Problem #16

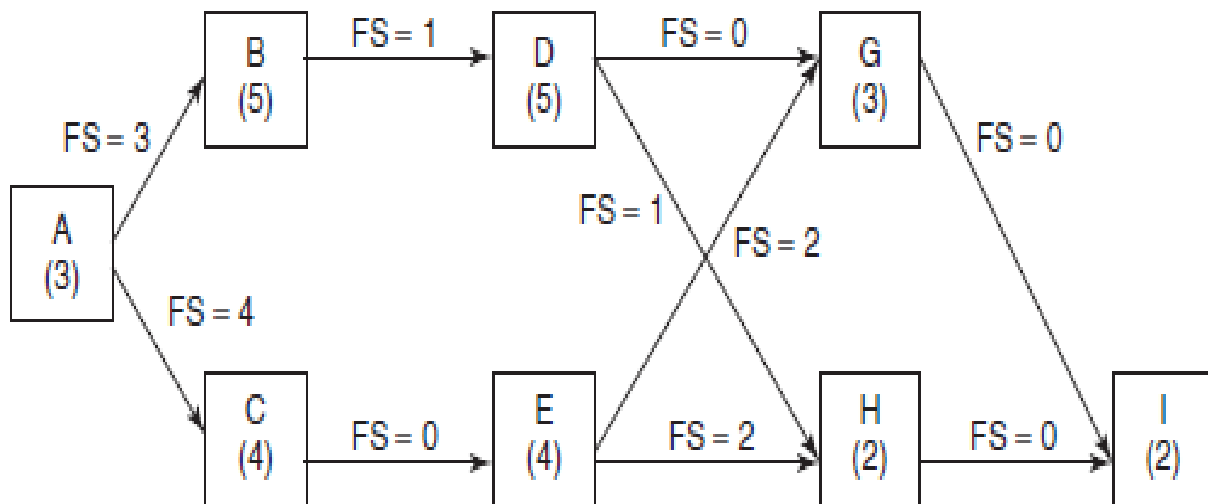


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For the given network, update the schedule of the project being done after 10 days, and the following observations are made on the progress of the activities:

- Activity A is complete.
- Activity B is complete.
- Activity C is complete.
- Activity D will take 3 more days to finish.
- Activity E still needs 4 days to complete.
- Activity F is underway and will take another day to complete.
- Duration of activity G has been revised to 8 days.

Problem #17



For the given network, update the schedule of the project being done at the end of 14 days, and the following observations are made:

- Activity A is complete.
- Activity B is complete.
- Activity C is complete.
- Activity D will take another five days to complete.
- Activity E is 50 per cent complete and will take another six days to complete.
- Activity G's duration has been revised to five days.
- Activity H's precedence relation has been updated to FS = 2 from activity D.
- Activity I will start three days after activity G.